MD MAZHARUL ISLAM EMON

Software Engineer | M.Sc. Computer Science student

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Date of birth: 26.07.1999 • **Nationality:** Bangladeshi

PROFILE

Software engineer with 3.5 years of production experience at LII Lab, now studying Computer Science at Paderborn University. Built Python backends and Flutter applications for IELTS and PTE preparation, including speech scoring, subscription billing and multi-tenant delivery. Recent independent work includes document retrieval, grounded LLM drafting and web products. Seeking a working-student role in software development, backend engineering or applied AI.

EDUCATION

Paderborn University — Paderborn, Germany

M.Sc. Computer Science — 10/2026 – present

Enrolled for Winter Semester 2026/27.

Leading University — Sylhet, Bangladesh

B.Sc. in Computer Science & Engineering - GPA: 3.6 / 4.0 — Degree awarded 2023

- **Relevant coursework:** Data Structures, Algorithms, Operating Systems, Computer Networks, Databases, Distributed Systems, Machine Learning.
- **Final-year project: Projecto:** multi-role university course & proposal management system built with Django, with role-based workflows, automated notifications, and PDF/CSV reporting.
- **Leadership:** Founding Chair, IEEE Computer Society LU Student Branch (2022–23); ACM Coordinator, LU Computer Club (2022–23).

LANGUAGES

Bengali: Native

English: IELTS Academic 6.5 (Listening 7.0 · Reading 6.5 · Writing 6.5 · Speaking 6.0), Aug 2025

German: No working proficiency

TECHNICAL SKILLS

Languages: Python · TypeScript · Dart · SQL · JavaScript · C++

Backend & APIs: Django · Django REST Framework · FastAPI · REST · WebSockets · Celery · async Python

Frontend & Mobile: Flutter · Riverpod · Next.js · React · Tailwind CSS · Zod

Data & Infra: PostgreSQL · Redis · Docker · Nginx · GCP · DigitalOcean · CI/CD · Prometheus · Grafana

AI / ML: OpenAI API · WhisperX · PyTorch · RAG · Speech Recognition · Pronunciation Assessment · NLP

PROFESSIONAL EXPERIENCE

LII Lab
Sylhet, Bangladesh

Senior Software Engineer
01/2025 – 04/2026

OnePTE & OneIELTS: AI-powered English test-prep SaaS (Web + iOS + Android).

- Architected **multi-tenant SaaS backend** with django-multitenant: full data isolation, custom-domain routing, per-tenant scoring config and feature flags; scaled to two production products on a shared codebase with zero cross-tenant data leakage.
- Designed a **QTI 3.0–compliant exam engine** supporting 100+ question types across IELTS Academic & General with XML parsing, interaction routing, and strategy-pattern band-score normalization.
- Built a **real-time AI speech-evaluation pipeline** (Speech Recognition + Pronunciation Assessment + NLP) delivering end-to-end scoring in under 15 seconds: combining WhisperX, OpenAI APIs, and custom NLP graders for fluency, lexical resource, and grammar.
- Engineered a **multi-gateway payment platform** across 5 providers (Stripe, Razorpay, SSLCommerz, Google Play, Apple) with subscription lifecycle management, webhook idempotency, and transaction-level reconciliation.
- Stood up a production **observability stack** with Prometheus + Grafana: health checks, per-API latency histograms, and error-rate alerting.
- Drove **AI-native engineering practice** across the team: Claude Code workflows, CLAUDE.md skill/subagent design, and AI-assisted PR review; mentored junior engineers through code reviews and architecture walkthroughs.

Stack: Django · DRF · FastAPI · Python · PostgreSQL · Redis · Celery · Docker · GCP · Prometheus · Grafana · Next.js · TypeScript

LII Lab
Sylhet, Bangladesh

Software Engineer
11/2022 – 12/2024

OnePTE: AI-scored PTE test-prep platform (Flutter + Django).

- **Full-stack ownership from day one**: designed and shipped the Flutter cross-platform app (iOS + Android) covering all four PTE modules and 20+ task types with audio capture, timer management, and offline-tolerant state via Riverpod.
- Built a **task-group mock-test engine** with modular exam templates, configurable time allocations, automated question progression, and multi-dimensional score breakdowns.
- Engineered the **AI scoring backend** for spoken and written PTE tasks using Speech Recognition, Pronunciation Assessment, and NLP for multi-trait evaluation.
- Integrated **subscription billing across 4 platforms** (Stripe, SSLCommerz, Google Play, Apple) with webhook handling, transaction deduplication, and in-app purchase receipt verification.
- Led **90+ production releases** across Android, iOS, and Web: owned release pipelines, Firebase Remote Config, and CI/CD across all channels.

Stack: Flutter · Dart · Riverpod · Django · DRF · Python · PostgreSQL · Redis · Celery · Firebase · Next.js · TypeScript

SELECTED PROJECTS

Sushi Lab: Bilingual Restaurant Ordering & Marketing Platform
2026

Japanese restaurant in Chartres, France: solo end-to-end build.

- Designed and shipped a production **bilingual (FR/EN)** ordering + marketing site on Next.js 16 App Router with React 19 and server actions: translated URL segments and locale-aware checkout.
- Built an **ordering engine** with a hydration-safe Zustand cart, scheduled-ahead time slots, per-item option pickers, and server-enforced time-of-day availability windows.
- Engineered **triple-channel notifications** (Telegram bot, transactional customer email, restaurant inbox) dispatched in parallel, plus AI-discoverable SEO (JSON-LD, per-locale canonical + hreflang) and a GA4 ecommerce funnel.

Stack: Next.js 16 · React 19 · TypeScript · Tailwind · Zustand · Zod · React Email · next-intl · ZeptoMail · Telegram Bot API · GA4

COMPETITIVE PROGRAMMING & ACHIEVEMENTS

Solved **2,000+ problems** across online judges; participated in **300+ contests** over 4 years.

Runner-up (Bangladesh), IEEEXtreme 16.0 (2022): global rank 149 / 6,500+ teams.

Codeforces (Max: **1603**) · CodeChef (Max: **1965**)

ICPC Dhaka Regional 2020 Main Round: 51st. **Regional Online Preliminary 2021**: 85th of 1,747 teams.

Champion, LU CSE Carnival National Hackathon 2023; **Champion**, LU TechStorm 4 (2021).